ALP-ttbar Toy MC v2.0: Series Overview (v1.0–v1.9) — What a Preregistered Toy Study Learned about Displaced Light ALP → e^+e^- and Photon Conversions, and Open Questions for Full-Simulation Studies

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No new simulation or fit: every number below is taken from the published records v1.0–v1.9 (list on the last page). Toy study motivated by Alimena et al., arXiv:2610.38932 [1], which extends Rygaard et al., JHEP 10 (2023) 138 [2].

Disclaimer. This is an independent, simplified, numpy-only toy study by a non-specialist. It is not affiliated with the authors of Refs. [1, 2] or with the CMS, LHCb or ALICE Collaborations or the FCC study. Kinematics are parametrised, the tracker is a set of idealised cylinders with an assumed material map, and the selection is optimised on the same samples. **Absolute numbers are not predictions for any real detector**; only trends and ratios inside the toy are meaningful. The open questions in §6 are offered as questions that full simulation could check, not as claims.

Summary. Over ten versions (eight of them preregistered) the series asked how photon conversions in tracker material limit a search for displaced light ALPs decaying to e^+e^- in $t\bar{t}$ events, and what reduces that limit. In this toy, at 17 MeV the loss in reach from the conversion veto ($c\tau$ = 1 mm) is ≈ 2× without bremsstrahlung but 126–178× with it at nominal material; better electron fits and a two-hypothesis vertex fit barely change it; **reducing the material is the only large handle** (127× → 35× for a quarter of the material), measuring z at every layer adds to it (→ 27×), and the **arrangement** of the pixel layers at fixed material does not matter. Of the 48 predictions frozen in v1.2–v1.9, 32 were confirmed and 16 refuted; the refuted ones, and one failed validation (Wilks' theorem in v1.5), taught the most. §6 lists six questions where the toy points but cannot decide, and §7 places the series next to published work on next-generation, low-material vertex detectors (ALICE ITS3, ALICE 3, FCC-ee).

1. The question and the two source papers

Rygaard et al. [2] proposed searching for long-lived ALPs produced with a top-quark pair through a displaced dimuon vertex: the reconstructed $t\bar{t}$ system provides trigger options and background suppression. For ALPs above the dimuon threshold they projected sensitivity to effective top couplings $|c_{tt}|/f_a$ as small as 0.03 (0.002)/TeV and proper decay lengths up to 10 (400) m for 150 fb^{-1} (3 ab^{-1}), assuming a 1 fb cross section. **Alimena et al. [1]** extended the strategy to a → e^+e^-, which reaches masses below the dimuon threshold: a CMS-motivated, generator-level study for 17 MeV–10 GeV and $c\tau$ = 0.1 mm–1 m, probing σ × B of order 10^{-3} pb with 500 fb^{-1} at 13 TeV. The authors state the limits of that projection explicitly: electron trajectories are approximated as straight lines, neglecting their curvature in the magnetic field; "backgrounds arising from detector-material interactions, object misreconstruction, and other instrumental effects have not been simulated"; and highly boosted low-mass ALPs "can also decay at large transverse displacements, reducing the tracker lever arm available after the decay" [1]. They also note that the opposite bending of electron and positron may increase their separation. Related experimental searches for collimated lepton pairs are cited there [23–25].

This series took exactly these stated limitations as its starting point. For the lightest ALPs the two electrons are almost collinear, and the displaced pair looks very much like a photon that converted in the tracker material. A standard analysis removes vertices near material ("material veto"); this also removes signal. The series asks, at toy level: **how large is this loss, which detector effects set it, and which handles reduce it?** The figure of merit throughout is the "limit degradation": the factor by which the expected limit worsens because of conversions, relative to a background-free search with the same selection. v1.0 addressed the field: with the 3.8 T bending, ≈ 97% of pairs become resolvable, consistent with the expectation stated in [1].

2. How the series worked

Toy ingredients. ALP and photon kinematics parametrised to match the published generator-level numbers of [1]; CMS-Phase-1-like barrel (four pixel and ten strip layers, 3.8 T), assumed material map (≈ 0.46 X_0 up to 800 mm at η = 0); from v1.3 a mini tracking simulation with exact helices, Highland multiple scattering [4] and Bethe–Heitler bremsstrahlung [3]; vertex fits by Gauss–Newton least squares; from v1.7 a longitudinal (r–z) simulation and fit. Everything is numpy, reproducible from fixed seeds (about 20–25 CPU hours for all versions together). **Preregistration.** From v1.2 on, each version froze its questions, method, sanity gate, validation criteria and numerical predictions on GitHub and Zenodo before any result was computed (README SHA-256 printed in each PDF). Deviations were reported, refuted predictions were not re-framed, and each version re-used the previous samples and checked bit-level identity (typically 100.000% of pairs) before changing one thing.

3. What happened, version by version

Fig. 14 The conversion-induced loss through the series (toy MC, 3.8 T; published numbers only; open marker = assumed, not simulated, resolution)

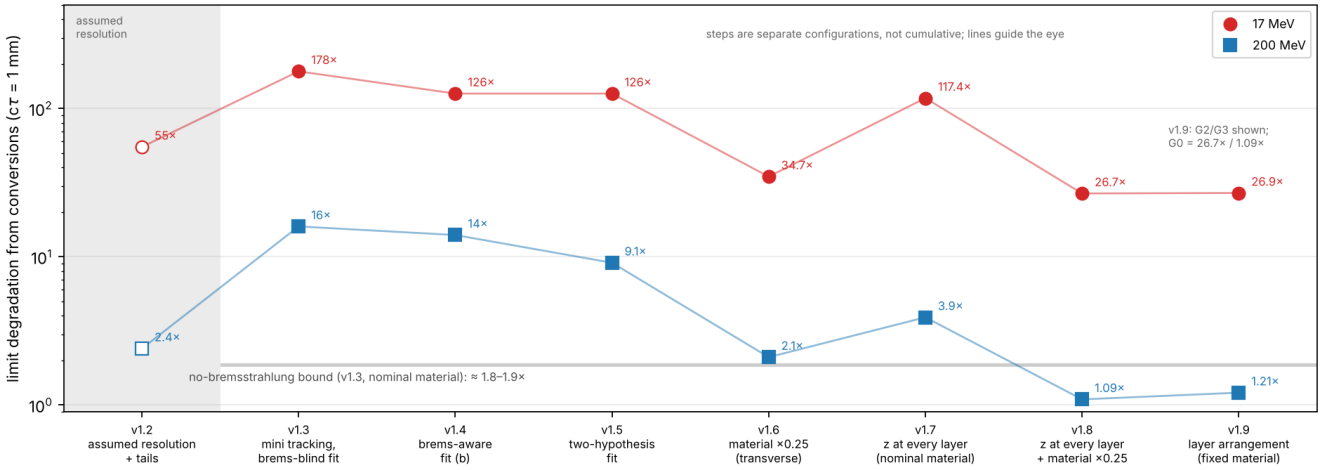


Figure 14. Limit degradation from conversions at 17 MeV and 200 MeV (cτ = 1 mm) for the main configuration of each version. The v1.2 point uses an assumed (not simulated) vertex resolution; the steps are separate configurations, not a cumulative sequence. Grey band: the v1.3 bound without bremsstrahlung.

version	what changed	17 MeV	200 MeV	main lesson
v1.0	B-field separation of collimated pairs; conversions as background	–	–	≈ 97% of pairs resolved at 3.8 T; conversions identified as the key background for the lightest ALPs
v1.1	per-pair vertex resolution; four resolution scenarios	1.8–64×	1.6–3.2×	the answer depends almost entirely on the vertex resolution and its tails
v1.2 P	hit-pattern consistency (assumed resolution + tails)	55×	2.4×	hit-pattern requirement helps less than predicted (64× → 55×; P1 refuted)
v1.3 P	mini tracking simulation; fit ignores brems	178×	16×	brems pushes conversion vertices outwards; without brems ≈ 1.9×
v1.4 P	brems-aware fits (inner hits; Gaussian brems noise)	126×	14×	calibration repaired (pulls 0.95), resolution not: information is lost
v1.5 P	two-hypothesis fit (conversion vs. ALP opening)	126×	9.1×	small openings are absorbed by the free vertex; Wilks fails (boundary)
v1.6 P	field (0.2–3.8 T) and material (×0.25–1) scan	35× (f = 0.25)	2.1×	material is the effective handle; lower field does not help 17 MeV
v1.7 P	adding the r–z opening (3D fit); stereo / all-pixel z	117× (all-pixel)	3.9×	r–z works from 50 MeV; at 17 MeV scattering fakes polar openings
v1.8 P	all-pixel z with reduced material; truth-momentum diagnostic	26.7×	1.09×	material and z coverage reinforce; 17 MeV limit = scattering, not brems
v1.9 P	layer arrangement at fixed material	26.7–26.9×	1.09–1.21 ×	arrangement hardly matters; more layers cost veto acceptance

Table 1. The series at a glance (P = preregistered). Degradations at cτ = 1 mm for the main configuration of each version; "–" = not computed in that form. Details in each record.

4. What was learned (within this toy)

1. **Bremsstrahlung, not the vertex hypothesis, sets the scale at nominal material.** Without bremsstrahlung the loss at 17 MeV is ≈ 1.9× (v1.3); with it, 126–178×. A fit that models the mean energy loss repairs the calibration of the vertex (bias +14 mm → +1.1 mm, pull width 11.9 → 0.95) but not its resolution (core ≈ 7–11 mm vs ≈ 2 mm), because the information carried away by the photon cannot be restored (v1.4).
2. **In the bending plane a free vertex hides a small opening angle.** Moving the vertex along the tracks reproduces an opening Δφ at first order; what remains scales as Δφ², below the hit resolution at 17–50 MeV (v1.5). The same singularity makes the null hypothesis a boundary point, so the likelihood ratio follows $½\chi^2_0 + ½\chi^2_1$, not χ^2_1 (Wilks [9] fails; Chernoff [10], Self–Liang [11]). The preregistered validation failed for this reason and was reported as such.
3. **The r–z view sees the opening at first order.** A vertex shift cannot absorb a polar opening and bremsstrahlung barely changes the polar angle, so with z measurements the separation works from 50 MeV upwards (v1.7). With z only in the pixel layers, few 17 MeV pairs (≈ 15%) have r–z information; with z everywhere, all do.
4. **At 17 MeV the opening is as small as the scattering.** The median polar opening of 17 MeV pairs is 0.42 mrad, comparable to the Highland angle of one pixel layer for a 10 GeV electron (0.16 mrad at nominal material). A diagnostic covariance that knows the true momentum after bremsstrahlung removes most fake openings of conversions (7.4% → 1.9%) but improves the 17 MeV loss only from 117× to 113× (v1.8):

scattering, not momentum mismodelling, is the limit there.

5. **The amount of material is the handle; its arrangement is not.** Scaling the material by f = 0.5 and 0.25 lowers the 17 MeV loss 127× → 65× → 35× (transverse, v1.6) and, with z at every layer, 117× → 56× → 27× (v1.8). At fixed material, a closer first layer or twice as many thinner layers changes it by ≤ 1% at 17 MeV (v1.9): the per-pair r–z gain (≈ 19% inside the pixel volume) applies to the quarter of 17 MeV pairs that decay there, and more layers mean more veto bands (signal removed 53% → 58%).
6. **The magnetic field is not a free lunch.** Lowering the field (3.8 → 0.2 T) improves the opening-angle test but degrades the vertex position of conversions (≈ 8 → 48 mm) and the mass resolution; at 17 MeV the loss rises (127× → 188×) (v1.6).
7. **Predicting got easier as understanding grew.** Confirmed predictions per version: v1.2 3/4, v1.3 1/5, v1.4 2/5, v1.5 3/6, v1.6 4/7, v1.7 6/7, v1.8 6/7, v1.9 7/7 (32 of 48). The largest miss came when realistic tracking was first simulated (v1.3: 183× against a predicted 20–150× at 17 MeV); later predictions, informed by the earlier failures and by analytic estimates (e.g. the Fisher-matrix estimate of v1.9), were mostly right.

5. What a toy can and cannot say

It can isolate one effect at a time on identical events, test the direction and rough size of an effect, and show which mechanism dominates (e.g. that the 17 MeV limit is scattering-, not fit-limited), with every step reproducible. **It cannot** give the absolute size of the loss in any real detector. The material map, band widths, the cylindrical barrel-only geometry, the absence of pattern recognition, pile-up, fake tracks, hadronic interactions, other backgrounds (K^0_S, Λ, Dalitz π^0 decays) and the simple counting rule all change the numbers; the selection is optimised on the same samples. The material scale in v1.6–v1.9 is a uniform factor: at f = 0.25 a toy pixel layer is still ≈ 0.5% X_0, several times thicker than next-generation designs (§7), and real low-material trackers redistribute supports and services rather than scaling them.

6. Open questions where full simulation could help

Each item states what the toy saw, why it may matter beyond this search, and what a full or fast simulation could check. They are offered as questions; the toy cannot answer them.

Q1. How much signal does the material veto itself cost for displaced e^+e^-?

Toy: before any exemption, the 2σ material veto removes 53–56% of base ALP pairs at f = 0.25 and 58–65% with eight thinner pixel layers (v1.9); in v1.9 this cost, not the resolution, decided the result. Check: the veto acceptance for displaced e^+e^- vertices with a realistic material map, and whether a three-dimensional or likelihood-based material-interaction tag (using z and the measured material structure) recovers signal. Relevance: any displaced-vertex search that vetoes material, including e^+e^- and hadronic LLP searches.

Q2. Is the 17 MeV limit scattering-dominated also in a real detector?

Toy: removing the momentum mismodelling (truth-momentum diagnostic) left the 17 MeV loss almost unchanged, while reducing the material lowered it by a factor ≈ 4 (v1.8). Check: in a full simulation, the separation of signal-like and conversion-like pairs versus material budget for sub-mrad openings, with and without a Gaussian-sum filter [5, 6]. Relevance: whether GSF-type electron fits or material reduction is the better investment for collimated e^+e^- pairs.

Q3. How useful is z information outside the pixel volume?

Toy: z at every layer roughly halves the loss at 50–200 MeV (v1.7), mainly because most light ALPs decay beyond the pixel layers (median decay radius 370 mm at 17 MeV, v1.9). Check: the value of z measurements in the outer tracker (stereo strips, pixelated outer layers, as in ALICE 3 [13] or FCC-ee concepts [15]) for vertices at 0.2–0.8 m. Relevance: outer-tracker technology choices for LLP physics.

Q4. Does the arrangement of thin layers matter less than their total material for displaced pairs?

Toy: at fixed material, moving the first layer from 29 to 20 mm or doubling the number of pixel layers changed the 17 MeV loss by ≤ 1% (v1.9). Note that this concerns vertices displaced by centimetres to tens of centimetres; for prompt impact-parameter resolution, first-layer radius and thickness are known to matter strongly [16, 18]. Check: the same comparison for displaced vertices with realistic geometries. Relevance: layout optimisation of next-generation vertex detectors beyond flavour tagging.

Q5. Which reference distribution does a two-hypothesis vertex test follow?

Toy: in the bending plane the opening-angle test is a boundary problem ($½\chi^2_0 + ½\chi^2_1$); in r–z it is regular (χ^2_1); their sum followed $½\chi^2_1 + ½\chi^2_2$ (v1.5, v1.7). Check: calibration of such tests with pseudo-experiments [12] before using fixed thresholds. Relevance: any conversion versus decay discrimination based on a likelihood ratio.

Q6. How large is the electron-specific penalty relative to muons?

Toy: without bremsstrahlung and conversions the loss is ≈ 2× (v1.3), compared with 27–178× for electrons. Check: a like-for-like electron versus muon comparison in the overlapping mass range (≈ 0.25–0.4 GeV), as done with real data for heavier LLPs by CMS [22]. Relevance: how much of the low-mass reach claimed for e^+e^- at generator level survives detector effects.

7. Related work: next-generation low-material vertex detectors (prior art)

The direction suggested by the toy (less material) is the established direction of vertex-detector R&D. The following public work was checked for this overview; it is not an exhaustive review.

- **ALICE ITS3** [14]: three truly cylindrical layers of wafer-scale, bent MAPS (≈ 30–50 µm thick), first layer at 18 mm around a 16 mm beam pipe, ≈ 0.05% X_0 per layer instead of ≈ 0.36% (ITS2); planned installation in 2027 for LHC Run 4; about a factor two better pointing resolution expected.
- **ALICE 3** [13]: an all-silicon tracker in a 2 T solenoid with a retractable vertex detector at ≈ 5 mm from the beam (≈ 0.1% X_0 per layer) and an outer tracker of ≈ 1% X_0 per layer, for LHC Run 5 and beyond.
- **FCC-ee vertex detectors** [15, 16]: in the IDEA concept the inner barrel layers sit at 13.7, 22.7 and 34 mm with ≈ 0.25% X_0 for the first layer and ≈ 2% X_0 for the whole vertex detector; an ultra-light concept with four layers of curved wafer-scale MAPS reaches ≈ 0.075% X_0 per layer and ≈ 0.35% X_0 in total, "about a factor three below the classic design" [16]. CLD uses double-sided layers, which helps at higher momenta while the lighter IDEA layout is better for 1 GeV tracks [16].
- **FCC-SEED** [17]: a French-led vertex-detector concept for FCC-ee based on bent ladders of large curved MAPS; MIMOSIS sensors 40–50 µm thick have been bent to 15–18 mm radius, with 12–15 mm and 30–50 µm planned; first studies in the CLD full simulation address how resolution, geometry and material affect tracking and impact parameter, and implementing such geometries in the Key4hep full simulation is part of current work.
- **Material vs. performance at FCC-ee** [18]: a fast-simulation study (IDEA, DELPHES) found that thinning the three innermost layers from 280 to 30 µm reduces the multiple-scattering term of the impact-parameter resolution by ≈ 30% and improves the secondary-vertex resolution by ≈ 20%; thinning only the first layer still gives ≈ 6%. An exploratory design study [19] states that "the material budget of the vertex detector is arguably as important a design parameter as its hit resolution" and targets ≈ 0.02–0.03% X_0 per layer.
- **Long-lived particles at FCC-ee** [20, 21]: displaced-vertex searches (heavy neutral leptons, ALPs, exotic Higgs decays) are a recognised physics case; the studies we checked use muon final states or fast simulation and do not model photon-conversion backgrounds to displaced e^+e^-, which is the topic of this series (in a different, LHC $t\bar{t}$ setting).

How the series relates. The toy reaches, from a different starting point (a specific LHC search with a CMS-like barrel), the same qualitative conclusion that drives these designs: for soft or collimated tracks, material dominates. It adds nothing to their engineering. What may be of interest is the angle: for *displaced* e^+e^- pairs, the veto acceptance (Q1), the outer-tracker z information (Q3) and the limited role of the layout (Q4) are quantities that the flavour-tagging-driven optimisations do not usually report. Even the reduced toy material (≈ 0.5% X_0 per pixel layer at f = 0.25) is several times above these designs; whether the trend continues there was not tested.

8. Limitations

Barrel only ($|\eta| < 1.2$); idealised cylindrical layers; assumed material map and band widths; no pattern recognition, fakes, pile-up or hadronic interactions; only photon conversions as background; single-Gaussian bremsstrahlung model in the transverse fit; sequential rather than joint 3D fit; truth used for the material list and scattering momentum in the covariance; signal kinematics parametrised; selection optimised on the same subsampled events; simple counting rule for the limit. All conclusions are statements about this toy.

9. Records

version	results (Zenodo DOI)	preregistration
v1.0 / v1.1	10.5281/zenodo.23825276 / .23833475	–
v1.2 / v1.3	10.5281/zenodo.23847880 / .23862775	GitHub (frozen README)
v1.4	10.5281/zenodo.23880375	GitHub (frozen README)
v1.5	10.5281/zenodo.23924862	10.5281/zenodo.23905511
v1.6	10.5281/zenodo.23952888	10.5281/zenodo.23942338
v1.7	10.5281/zenodo.24011489	10.5281/zenodo.23995668
v1.8	10.5281/zenodo.24034481	10.5281/zenodo.24028384
v1.9	10.5281/zenodo.24052153	10.5281/zenodo.24048262

Code for all versions: github.com/wspaik5 (latest complete repository: ALP_ttbar_Toy_MC_v1.9_Postregistration_Layer_Geometry). The one-page overviews posted with the compilations of v1.0–v1.2, v1.6–v1.7, v1.8 and v1.9 are included unchanged in the folder linkedin_overviews/ of this record.

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